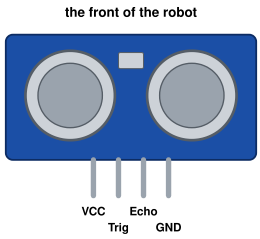
How to plug in the distance sensor

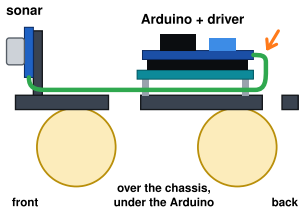
Then use it in Blockly: if it sees something, do this, else do that

1. Find the sonar



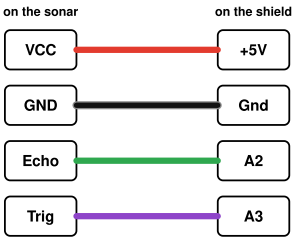
The **sonar** is the pair of round **"eyes"** at the front of the robot. Its **4 pins** are labeled **VCC**, **Trig**, **Echo**, **GND**.

2. Follow the wires



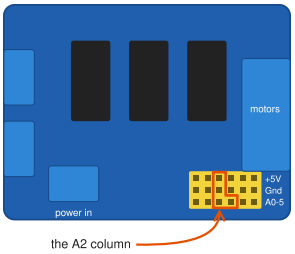
Look from the side. The four wires start **just behind the sonar**, run back **over the chassis**, **under the Arduino**, and loop up into the **driver board** at the back of the tower.

3. Which wire is which



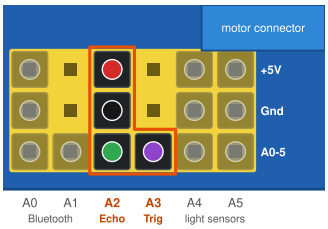
VCC goes to **+5V**, **GND** to **Gnd**, **Echo** to **A2**, **Trig** to **A3**. Follow each wire from the sonar back to the shield. Your wires can be any colors.

4. Find the analog header



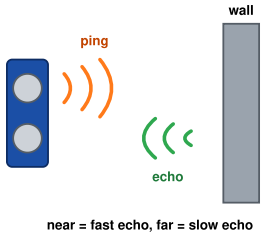
The other ends go on the motor shield, on the analog header: the **top row is all +5V**, the **middle row is all Gnd**, the **bottom row is A0 to A5**. The sonar uses the **third column, A2**.

5. Make an L



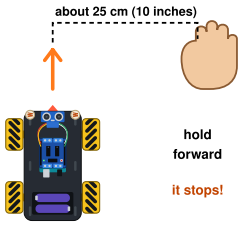
The four wires make an **L**: **+5V**, **Gnd** and **A2** (Echo) down the A2 column, and **A3** (Trig) next to A2. It sits right next to the Bluetooth L.

6. How it sees



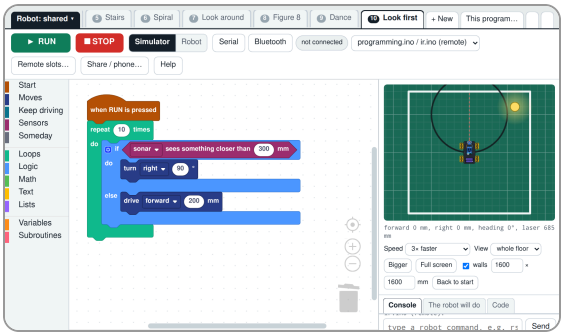
Trig sends a **ping** of sound too high to hear. Echo listens for it to bounce back: the longer the wait, the **farther away**. Soft or thin things are hard to see.

7. Check it



Hold **forward** on the remote, hand in front: the robot **stops about 25 cm away**. (Robot program **0.10.87** or newer.)

8. Program it: if ... else



Blockly

robotics.makerlabkids.com/mecanum/blocks

In **Sensors**, find **sonar sees something closer than 300 mm**. Put it in an **if ... else**: if it sees something, turn; else, go forward. Or press **+ New** and pick **Look first**. Connect with **Bluetooth**, choose **Robot**, press **RUN**. Programs with **if** run from the computer, not from a remote slot.